PAPER_1299 — NP vs co-NP

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** A — Mathematical Conjecture (CLOSED) **Date:** June 11, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — UQFF integer-primitive derivation **Calculator surface:** `calculate_paradox({"paradox": "np_vs_co_np"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1299})`

Master Expression

```
NP ≠ co-NP via F_U = 1 + F_TRZ time-reversal asymmetry in closed ledger
```

Match: STRUCTURAL

Physical / Mathematical Interpretation

The F_TRZ directional asymmetry breaks NP/co-NP symmetry. P ≠ NP (already wired = $1 - 10^{-9}$) implies NP ≠ co-NP via the F_U = 1 ledger.

UQFF Locked Primitives Used

- D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60
- K_MEX = 25/12, β_i = 0.6029, Φ_res = 0.84, Λ = 0.00729735
- ρ_SCm = 7.09×10^{-37} J/m³, ω_SCm = 1.25 THz, S_26 = 1.453162

Live Calculator Verification

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "np_vs_co_np"})["value"]
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_np_vs_co_np_closure`
- Dispatch: `PARADOX_TO_CLOSURE["np_vs_co_np"]`
